

| Question |     |       | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                        | Marks available |            |     |       |       |      |
|----------|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------|-----|-------|-------|------|
|          |     |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                        | AO1             | AO2        | AO3 | Total | Maths | Prac |
| 7        | (a) |       | <u>Continuous spectrum</u> due to radiation of all wavelengths emitted from surface of star [1]<br>[superimposed] line <u>absorption spectrum</u> (due to passage of radiation) <u>through atmosphere</u> (of star) [1]                                                                                                                                                                                                                                | 2               |            |     | 2     |       |      |
|          | (b) | (i)   | Use of $I = \frac{P}{4\pi R^2}$ [1]<br>Substitution:<br>$\frac{I_S}{I_V} = \frac{9.7 \times 10^{27} \times [4\pi] (2.4 \times 10^{17})^2}{1.5 \times 10^{28} \times [4\pi] (8.1 \times 10^{16})^2}$ [1]<br><br><b>Alternative for 2nd mark:</b><br>Calculation of either intensity:<br>$I_{\text{Sirius}} = 1.18 \times 10^{-7} \text{ [W m}^{-2}\text{]}$ or $I_{\text{Vega}} = 2.07 \times 10^{-8} \text{ [W m}^{-2}\text{]}$<br><br>Ratio = 5.7 [1] | 1               | 1<br><br>1 |     | 3     | 2     |      |
|          |     | (ii)  | Shape of curve below that of Sirius at all points [1]<br>Peak at $(1, \lambda_{\text{max}})$ <b>ecf</b> [1]                                                                                                                                                                                                                                                                                                                                            |                 | 2          |     | 2     |       |      |
|          |     | (iii) | Substitution into $P=4\pi R^2 \sigma T^4$ i.e.<br>$9.7 \times 10^{27} = 1.8 \times 10^{19} \times 5.67 \times 10^{-8} \times T^4$ [1]<br>$T = 9.9 \times 10^3 \text{ K (9874 K)}$ [1]<br><br>Substitution into $\lambda_{\text{max}} = \frac{W}{T} = \frac{2.90 \times 10^{-3} \text{ [m K]}}{9.9 \times 10^3 \text{ [K]}}$ <b>ecf</b> [1]<br><br>$\lambda_{\text{max}} = 2.9 \times 10^{-7} \text{ [m]}$ [1]                                          | 1<br><br>1      | 1<br><br>1 |     | 4     | 4     |      |

| Question |     |  | Marking details                                                                                                                                                                                                                                                                                                                                                           | Marks available |          |          |           |          |          |
|----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|----------|
|          |     |  |                                                                                                                                                                                                                                                                                                                                                                           | AO1             | AO2      | AO3      | Total     | Maths    | Prac     |
|          | (c) |  | Reference to multiwavelength astronomy / using different parts of the em spectrum [1]<br>Early photographs used visible light so limited / extra wavelengths <b>provide extra information</b> [1]<br>Extra detail provided e.g. link between wavelength and temperature such as quasars at gamma / X-rays [1] This detail could be evidence for the 2 <sup>nd</sup> mark. |                 |          | 3        | 3         |          |          |
|          |     |  | <b>Question 7 total</b>                                                                                                                                                                                                                                                                                                                                                   | <b>5</b>        | <b>6</b> | <b>3</b> | <b>14</b> | <b>6</b> | <b>0</b> |